

Formal Runtime Verification for Tool-Using LLM Agents: An Offline Same-Benchmark Study on AgentDojo and STAC

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ABSTRACT

Guardrails for tool-using LLM agents are often *rule-based*: a mostly per-call predicate decides each tool call, written per application without a common formal semantics, so multi-step, data-dependent policies are hard to state, audit, and reuse. Metric first-order temporal logic (MFOTL) and its monitor **MonPoly** offer a declarative alternative for quantified temporal policies over a tool-call *trace*. We ask, empirically and offline, how far this formalism reaches on recorded agent-attack corpora. We express five generic obligations in MFOTL and replay three benchmarks of recorded trajectories — AgentDojo, STAC, and R-Judge — with the real MonPoly engine (no LLM, no API key). RV flags 71.8% of STAC chains and detects 61.6% of GPT-4o’s *successful* AgentDojo attacks at a 31.7% benign firing rate (BFR); we are explicit that here the generic obligations reduce largely to *typed-action detection*, since approvals are almost never present. The formalism’s distinctive value appears with *parametric* specifications that quantify over trace data: a payee-provenance obligation cuts banking BFR from 44% to 24% and an external-recipient obligation cuts workspace BFR from 15% to 11%, both statistically significant (exact McNemar $p < 10^{-10}$). We further show an earliest-warning margin (obligations fire before the final harmful call in 51.3% of successful attacks), that the provenance predicate is itself an attack surface, and a timed CPS/IoT case study in which *elapsed time* alone flips MonPoly’s verdict on the same building-actuation action. We position MFOTL/MonPoly as a reusable, auditable substrate for agent-trace policies — complementary to, not a replacement for, rule-based enforcement.

CCS CONCEPTS

• **Security and privacy** → **Software and application security**; • **Theory of computation** → *Logic and verification*; • **Computing methodologies** → *Intelligent agents*.

KEYWORDS

runtime verification, LLM agents, CPS/IoT security, tool-call monitoring, MonPoly, past-time temporal logic, agent guardrails

1 INTRODUCTION

The defences shipped with tool-using large language model (LLM) agents are, in practice, mostly *rule-based*. AgentSpec [15], Progent [14] and GuardAgent [16] evaluate predicates on tool calls; the defences benchmarked in AgentDojo [6] (tool filters, prompt spotlighting, prompt repetition) are per-call or per-prompt heuristics. Some of these engines maintain state, and with enough bespoke code a stateful rule can encode a finite-history check — so we do *not* claim rules are incapable of history-dependent enforcement. What they typically lack is a *common declarative semantics* for the multi-step, data-dependent policies that the safety-relevant attacks exploit: each policy is re-encoded per

application, in imperative code, hard to audit, reuse, or verify. And those attacks are multi-step: STAC reports attack-success above 90% precisely because the violation is assembled across a chain of individually-benign calls [11].

Monitoring a *trace* against a declarative temporal specification is the defining task of runtime verification. First-order temporal monitors — MonPoly [4], DejaVu [8] — decide metric, quantified past-time properties over event streams with data, which is the exact shape of an agent obligation (“ $\forall$ object o : an irreversible action on o requires a prior approval for o ”). This makes metric first-order temporal logic (MFOTL) and its monitor MonPoly a natural *declarative substrate* for such policies: quantified over data, with a formal semantics and a reusable, independently-maintained engine, rather than bespoke per-application code — complementary to rule-based enforcement, not a claim that rules cannot do it.

The empirical question. A paradigm argument is only as good as its evidence: does formal RV actually catch *real* agent attacks, and what does it cost? Answering usually needs a live agent and an LLM. We avoid that: three major benchmarks ship *recorded* trajectories — AgentDojo publishes 36.7k recorded runs (tool-call traces plus its own attack-success and utility labels), STAC publishes 483 recorded attack chains, and R-Judge [17] ships labelled agent records with an IoT subset. We can therefore run a real RV engine over the *same* benchmarks the community uses, entirely offline.

Contributions.

- A framing of quantified temporal RV (MFOTL/MonPoly) as a declarative, reusable substrate for agent-trace policies — complementary to rule-based enforcement — and a library of five generic obligations evaluated by the **MonPoly** engine (§3), with an offline same-benchmark pipeline (§4).
- Results on three corpora with Wilson 95% intervals: 71.8% of STAC chains flagged (MonPoly-confirmed); 61.6% of GPT-4o’s *successful* AgentDojo attacks detected at a 31.7% benign firing rate (BFR); the full R-Judge (571 records) bounds the structural reach (§5).
- Two *parametric* obligations, where the verdict depends on genuine relational reasoning over trace data: a payee-provenance obligation (banking BFR 44% → 24%) and an external-recipient obligation (workspace BFR 15% → 11%), not stateless per-call predicates; pooled across all model directories the reductions are statistically significant (exact McNemar $p < 10^{-10}$; §6).
- A CPS/IoT case study answering the *temporal-complexity* question head-on: *timed* (metric MFOTL) building-actuation policies — bounded consent, sliding-window rate limit, timed precedence — run through MonPoly, whose verdict flips on identical actions purely by *elapsed*

time (§9); plus a canonicalisation error analysis over 54 hand-labelled tool→class mappings (§7).

Research questions. The study is organised around five questions. **RQ1** (coverage): how many recorded attacks do the obligations flag (§5, Tables 1–2)? **RQ2** (benign impact): what benign firing rate does that coverage cost (§5, §6)? **RQ3** (parametric precision): do data-parametric specifications improve discrimination over generic ones (§6, Table 3)? **RQ4** (temporal/metric expressiveness): are there policies whose verdict depends on elapsed time (§9, Table 4)? **RQ5** (robustness and practicality): how sensitive is the result to canonicalisation, how fast is the engine, and is the provenance signal itself attackable (§7, §8)? We also report an earliest-warning margin (§5) and a benchmark readiness scorecard (§10).

Scope. We measure whether an obligation *fires on a recorded trajectory* (detection under post-hoc replay), not live-agent attack success under an in-loop shield. Studying an in-loop shield addresses a *different* question — agent adaptation after enforcement — outside the observational scope of this work.

2 BACKGROUND: AGENT GUARDRAILS AND RV

Rule-based agent guardrails. A per-call guardrail evaluates a propositional predicate on the pending tool call. AgentSpec [15] and Progent [14] give rule languages over the current call; GuardAgent [16] emits guard code from a knowledge base; AgentDojo’s built-in defences [6] filter tools or rewrite prompts; Llama Guard [9] classifies message content. These are effective on their target threats but reason over the current call (or prompt), not the *quantified past* of the trace, and each policy is re-encoded per application.

Benchmarks with recorded traces. AgentDojo [6] evaluates prompt-injection attacks and defences across four suites (banking, slack, travel, workspace) and *ships* the resulting runs: per (model, suite, user task, attack, injection) a JSON with the assistant tool-call trace and the labels security (the injected task was accomplished) and utility (the user task was accomplished). STAC [11] ships 483 recorded multi-turn attack chains (derived from SHADE-Arena [10] and Agent-SafetyBench [18]), each a sequence of benign calls plus a withheld final harmful step. R-Judge [17] ships labelled agent records across domains, including a smart-home / traffic / phone IoT subset. All are replayable offline.

First-order temporal RV. MonPoly [4] monitors MFOTL over timed event streams, quantifying over data; DeJaVu [8] is a first-order past-time monitor and VeriMon is MonPoly’s formally verified core — all drop-in for the past-time obligations here. Stream monitors (RTLola [7], TeSSLa [5]) target numeric signals and complement us for quantitative obligations. A *hybrid* deployment can route relational/parametric obligations to MonPoly and purely numeric ones to RTLola/TeSSLa, reusing hybrid-hierarchical RV infrastructure that already runs both families in edge-IoT settings [2, 3]; a shield [1] gates the pending call on a violation.

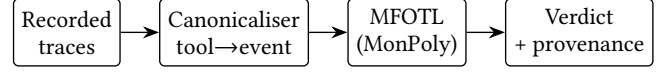


Figure 1: Offline pipeline. Recorded AgentDojo/STAC/R-Judge traces are canonicalised to a typed event stream and checked against MFOTL obligations by MonPoly; verdicts with provenance drive the offline analyses (coverage, BFR, earliest-warning, provenance integrity). No LLM or agent is executed.

3 FIVE GENERIC OBLIGATIONS IN MFOTL

We define five reusable obligations, each of the form “a risky action of a given class requires a matching precondition (approval/consent/confirmation) earlier in the trace,” chosen by action *semantics* rather than any benchmark’s tool names: **O1** irreversible/destructive (delete, cancel, purge), **O2** protection teardown (disable, unlock, add auto-delete rule), **O3** financial commit (transfer, pay, schedule transaction), **O4** external export (send, share, upload), and **O5** actuation/config (set, execute, adjust a device). In MFOTL (O1 shown; $\blacklozenge$ is once-in-the-past):

$$\Box \forall a. (\text{destructive}(a) \rightarrow \blacklozenge \text{approved}(a)).$$

These are the specifications we hand to MonPoly (as their monitored violation form; full formulas and keyword patterns in Appendix A). They are deliberately generic; the study measures how far five reusable formal templates go on real benchmarks, and where precision then demands *parametric* refinement (binding the approval to the specific object, within a window) — expressible in MFOTL but not in stateless per-call predicates.

4 OFFLINE SAME-BENCHMARK METHODOLOGY

Figure 1 shows the offline pipeline.

Canonicalisation. For each recorded run we build one event stream from the assistant tool calls (AgentDojo: {function, args} per call; STAC: the interaction history plus the withheld final step; R-Judge: the agent-turn actions). The monitor types each tool into an obligation class by semantic keyword patterns and detects preconditions by approval patterns (Appendix A).

Real engine. For STAC we additionally export the canonical stream to a MonPoly log and the obligations to MFOTL, and run the *unmodified MonPoly* binary; its firings are the RV result, not a re-implementation. The CPS/IoT policies of §9 are likewise evaluated by MonPoly.

Measures. On AgentDojo we use its labels: over attacked runs, *detection on successful attacks* (of security=true runs, the fraction where an obligation fires) and the *benign firing rate* (BFR: of benign runs, the fraction where an obligation fires). BFR is a potential-block proxy for the utility cost — what a shield *would* gate — not a measured task-failure rate, which would require re-running the agent. On STAC (attack-only) we report *final-step detection* (fires on the withheld harmful step); on R-Judge, detection on unsafe records and false-positive rate on safe ones. Confidence intervals are Wilson 95% score intervals.

Table 1: AgentDojo recorded runs (GPT-4o, no defence), replayed offline through our MFOTL obligations. Detection is on AgentDojo’s security=true (successful) attacks; BFR is on its benign (attack=none) runs. Per-obligation breakdown in Appendix B, Table 7.

Measure	value
Attacked runs	6680
AgentDojo attack-success (no defence)	1931/6680 = 28.9%
RV detection on <i>successful</i> attacks	1190/1931 = 61.6%
RV fires on any attacked run	2426/6680 = 36.3%
RV BFR on benign runs	39/123 = 31.7%

Benign sampling and parameter choice. The unit is one recorded run. A benign run is one user task with *no* injection (≈ 30 per suite, so 123 benign GPT-4o runs with a parseable trace); an attacked run is a (user task $\times$ injection $\times$ attack) combination (6680). The benign set is thus small, and we report Wilson intervals on every rate (e.g. the 31.7% aggregate BFR has interval [24.1, 40.4]). The keyword patterns are the ordinary English verbs for each class (Appendix A), fixed *before* measuring outcomes and never adjusted; the one threshold, the $\diamond$ window, is the most permissive choice (unbounded past), so a firing is never a tight-window artefact. “Detection” means the obligation fires on the recorded call; a would-be shield gating it assumes prevention. STAC’s content-semantic failure modes (harmful free text) are out of structural scope and flagged, if at all, only via the delivery act.

5 RESULTS

AgentDojo (same-benchmark, offline). Table 1 replays all recorded GPT-4o runs (no defence). AgentDojo’s own no-defence attack-success is 28.9% (1931/6680). Our RV, applied to the identical runs, fires on 61.6% (95% CI [59.4, 63.8]) of the *successful* attacks — i.e. a shield would have had a flagged call to gate in nearly two-thirds of the attacks that actually got through — driven mostly by the financial, export and actuation obligations (teardown never fires: these suites have no teardown tools). The cost is real and we report it plainly: the same generic obligations fire on 31.7% ([24.1, 40.4]) of *benign* task runs, because a legitimate task also sends email, moves money and changes config. Detection and BFR rise and fall together under *generic* specs.

Across models (RQ1–RQ2). The pattern is not GPT-4o-specific: across the 22 base (no-defence) model directories AgentDojo ships, RV detection on successful attacks spans 40.9–83.3% and BFR 17.6–37.4%. AgentDojo’s own built-in defences are not comparable to our offline replay, since they execute the agent *with* the defence in the loop, whereas we never re-run the agent.

Earliest warning: when the obligation fires (prevention timing). A detection is only useful for prevention if it arrives before the harmful step. On the 1931 successful GPT-4o AgentDojo attacks, an obligation fires *before* the final harmful call (*preventive*) in 991 (51.3%), fires only *at* that call (*last-moment*) in 199 (10.3%), and never fires in 741 (38.4%); where it is preventive the median detection lead is two tool-calls (mean 2.41, max 13). On STAC the picture inverts by

Table 2: STAC recorded chains (483) replayed through the same obligations; final-step detection. The MonPoly engine reproduces the 347 firings exactly.

Measure	value
Final-step detection (ref. monitor)	347/483 = 71.8%
confirmed by MonPoly	347 (exact)
structural-intent modes	249/342 = 72.8%
Out-of-template misses	136
Content-semantic (needs semantics)	141 ($\approx 29\%$)

construction — the corpus *withholds* the final harmful step, so the signal concentrates there: 76/483 (15.7%) preventive, 278 (57.6%) last-moment, 129 (26.7%) no firing. The earliest-warning margin, not a post-hoc residual, is our intervention-timing evidence.

Per suite: where generic obligations break down. The suite breakdown (Appendix B, Table 8) exposes the precision problem concretely. In **banking**, every action is a financial commit, so the generic obligations flag 97.7% of attacks *and* 95.8% of benign runs — near-total recall at near-total BFR. Travel, with few money/export actions, is the opposite (16.5% detection, 11.1% BFR). No single generic threshold is right for all suites; the domain-specific *parametric* refinements (§6) are what recover precision.

STAC (recorded chains, MonPoly). Table 2 replays all 483 STAC chains. Our reference monitor flags the withheld harmful action in 347/483 = 71.8% (95% CI [67.7, 75.7]) of chains, and the *unmodified MonPoly engine*, fed the MFOTL obligations and the exported log, fires on *exactly* the same 347 — the engine confirms the result. On the structurally-intended failure modes the rate is 72.8%; the content-semantic modes (harmful text, trusting/filtering tool results, $\approx 29\%$ of cases) are out of structural scope and flagged only via the delivery act. 136 chains end in an action outside the five generic classes (spam-rule edits, calendar changes, in-place appends), mapping where new obligations or semantics are needed.

R-Judge: the structural limit on a broad safety benchmark. To probe where structural RV runs out, we replay the *full* R-Judge [17] — 571 records (301 unsafe, 270 safe), a genuine benign set unlike STAC. The generic obligations fire on 23.6% (71/301) of unsafe records at 16.3% (44/270) FPR. The low detection is expected: most R-Judge unsafe cases are *content-semantic* (privacy leakage, misinformation, unsafe advice) with no structural trace signature, so a rule-based per-call checker fares no better. R-Judge thus *bounds* the reach of structural trace-level RV — formal or rule-based alike — and confirms a semantic aspect is the necessary complement.

6 PARAMETRIC OBLIGATIONS: THE CENTREPIECE

The AgentDojo BFR is not an artefact of RV — it afflicts *any* action-type checker, rule-based or formal, because “sends email” is both the attack and the job. Closing it needs a *precise* condition that quantifies over trace data: allow the transfer only if a matching approval for *that* payee occurred, or the export only if the recipient came from the user’s own directory. Such conditions are expressible in MFOTL and monitorable by MonPoly, and are *not expressible*

as a *stateless per-call predicate*, which cannot refer to the payee’s past provenance. We are careful here: a *stateful* rule engine could of course compute the same condition procedurally, by accumulating seen payees in bespoke code. The advantage we claim is therefore not exclusive computability but *declarative specification* — one formula, with formal temporal/relational semantics, evaluated by a reusable, independently-maintained engine, and hence auditable — rather than per-application imperative state management. We implement two and measure them (Table 3).

Payee provenance (banking). Instead of firing on *any* transfer (generic O3), the parametric obligation fires only on a transfer to a payee never seen as a *structured* recipient/sender in prior account data,

$$\Box \forall p. (\text{transfer}(p) \wedge \neg \blacklozenge \text{known_payee}(p)).$$

On GPT-4o banking runs this cuts the benign firing rate from 44.0% to 24.0% while detection on successful attacks falls only from 79.2% to 70.0%: the injected attacker IBAN is embedded in free text and is never a real counterparty, so it is still flagged, whereas a legitimate transfer to a known payee no longer is.

External recipient (workspace). Symmetrically for export, the parametric obligation fires only on a `send_email` whose recipient never appeared as a structured directory entry (contact/participant/sender field) in a prior tool result,

$$\Box \forall r. (\text{send}(r) \wedge \neg \blacklozenge \text{known_contact}(r)).$$

The injection payload delivers the attacker address inside a free-text email body, not the directory, so it is treated as external; a colleague already in the calendar or address book is not. This cuts the workspace BFR from 15.2% to 10.9% at a detection cost of 42.5% $\rightarrow$ 39.2%.

Discrimination (RQ3). The positive likelihood ratio LR^+ (detection / BFR) summarises both error rates in one number. Refining the generic obligations ($\text{LR}^+ = 1.94$) to the parametric spec raises it on both suites — payee provenance $1.80 \rightarrow 2.92$, external recipient $2.79 \rightarrow 3.61$ — so the parametric conjunct separates benign from malicious transfers/exports better than typed-action detection.

Statistical significance. The parametric obligation is a strict refinement of the generic one, so the paired comparison is concordant except for the discordant cell where the generic fires but the parametric does not. Single-model denominators are too small to be conclusive; pooling benign runs across all AgentDojo model directories (700 banking, 1,376 workspace) makes every discordant run a reduction (banking 75 vs 0, workspace 37 vs 0), with detection largely preserved. The exact McNemar test is then $p = 5.3 \times 10^{-23}$ (banking) and $p = 1.5 \times 10^{-11}$ (workspace).

What depends on temporal/parametric reasoning. We are explicit about the provenance of each number. The generic O1–O5 detection and BFR rates (Tables 1–2) are *typed-action detection*: a firing depends only on the tool’s class, not on trace history, because approvals are almost never present in these corpora. The two parametric rows in Table 3, and every verdict flip in the CPS/IoT study (§9), *do* depend on genuine relational or temporal reasoning: the *same* call is allowed or flagged according to a prior-provenance or prior-state condition — which a *stateless* per-call predicate cannot

Table 3: Parametric vs. generic obligations (GPT-4o), with Wilson 95% intervals. The parametric rows fire only when a relational/provenance condition over the trace holds, so their verdict differs from the same call’s typed-action verdict; the generic detection numbers elsewhere are typed-action detection.

Suite	Spec	detect % [CI]	BFR % [CI]
banking	generic O3	79.2 [75.7, 82.3]	44.0 [26.7, 62.9]
banking	param. payee	70.0 [66.1, 73.6]	24.0 [11.5, 43.4]
workspace	generic O4	42.5 [38.3, 46.8]	15.2 [7.6, 28.2]
workspace	param. ext-recipient	39.2 [35.1, 43.5]	10.9 [4.7, 23.0]

decide, and which MFOTL states declaratively rather than leaving to bespoke stateful code.

7 CANONICALISATION: ACCURACY AND SENSITIVITY (RQ5)

Detection is only as good as the tool $\rightarrow$ class typing. A single annotator, blind to the typing predictions and to any detection outcome, hand-labelled 54 distinct tool names sampled stratified by call frequency and domain across AgentDojo and STAC. Against this gold set the keyword typing has micro-averaged precision and recall both 0.84 and single-primary exact-match 83.3% (45/54); the residual errors are systematic (e.g. `send_channel_message` missed by export, `read-only get_*_transactions` false-typed financial) rather than random, and we report rather than tune them (per-class table in Appendix B, Table 9). To bound how much this typing drives the headline numbers, we re-ran the replays under four tool $\rightarrow$ class mappings (original, corrected, conservative, expanded; Table 10): AgentDojo detection moves in [61.6, 71.2]% and BFR in [31.7, 45.5]%, STAC detection stays in [61.7, 73.1]% — moderately sensitive, with the frozen typing *conservative* on AgentDojo (correcting the mapping raises detection, not lowers it).

8 PROVENANCE INTEGRITY: THE MONITOR’S TRACE HISTORY IS AN ATTACK SURFACE

The parametric obligations of §6 draw their power from a *provenance* predicate — `known_payee`, `known_contact` — computed over the monitor’s own recorded trace: a payee is trusted iff it appeared earlier as a structured recipient/sender field. That trace is, however, partly attacker-controlled: an injection can plant text that the canonicaliser reads as a structured directory entry, so the history the monitor trusts is itself an attack surface (RQ5).

Poisoning the provenance signal. We test this directly, injecting one line naming the attacker recipient into each flagged run’s readable context in four framings. A *structured* directory line or a *quoted/forwarded* line flips 94.0% of the 403 flagged banking-payee runs and 99.0% of the 203 flagged workspace-recipient runs to “known” (no longer flagged); a *free-text* mention and a *reformatted* recipient flip 0% (per-framing counts in Appendix B). One planted structured or quoted line therefore defeats the provenance check for 94–99% of the transfers it would otherwise flag: the check trusts *syntactic* structure it cannot authenticate.

Table 4: Timed (metric MFOTL) policies evaluated verbatim by MonPoly. For each, the *same* action is allowed or a violation purely by elapsed time; • marks the timepoints where MonPoly fires. Full sig/log/formula in Appendix A.

Policy (window)	action, timing	MonPoly
T1 consent (24 h)	export, consent 12 h ago	pass
T1 consent (24 h)	export, consent 48 h ago	• violation
T1 consent (24 h)	export, re-consent 10 h ago	pass
T2 rate ($\geq 3/6$ h)	3rd actuation within 6 h	• violation
T2 rate ($\geq 3/6$ h)	actuactions spaced > 6 h apart	pass
T3 precede (8 h)	execute, auth 6 h ago	pass
T3 precede (8 h)	execute, auth 40 h ago	• violation
T3 precede (8 h)	execute, re-auth 4 h ago	pass

A *hardened provenance specification*. The fix is to make provenance integrity part of the specification. A hardened obligation should require that (i) provenance come from a *trusted tool result* (an account/directory API response), not message text; (ii) a trusted contact *predate the session*; (iii) an approval *bind to the exact payee and amount*, not the action class; and (iv) a *new payee* require explicit confirmation. Conditions (i)–(iv) are themselves MFOTL obligations over a suitably-typed trace (a trust-domain field, a session id, an approval bound to object and amount) — exactly the schema current benchmarks lack (§10). Provenance integrity is a first-class requirement for agent RV, not a canonicalisation detail.

9 CPS/IOT CASE STUDY: TIMED TEMPORAL POLICIES

This is a proof of *applicability* of MFOTL/MonPoly to CPS/IoT policies (RQ4), not a broad security validation on CPS/IoT deployments. We give three evidence points, all evaluated verbatim by MonPoly: *timed* (metric) actuation policies, a multi-instance building trace, and a real (small) IoT subset.

Timed policies whose verdict depends on elapsed time. A reviewer of an earlier version noted the generic obligations exercise little temporal complexity — on these corpora an unbounded $\diamond$ reduces to typed-action detection. We therefore author three *metric* obligations whose verdict on the *identical* action changes purely with the clock (signatures, logs and formulas in Appendix A; Table 4). (T1) *Bounded-window consent*: an export requires a consent within the last $W=24$ h, $\text{export}(o) \wedge \neg(\text{ONCE}[0, 24] \text{ consent}(o))$ — MonPoly fires only when the consent is stale. (T2) *Sliding-window rate limit*: ≥ 3 actuactions on a device within $W=6$ h (metric CNT over $\text{ONCE}[0, 6]$) — fires on a burst, silent once calls are spaced. (T3) *Timed precedence*: an execute must be preceded within $W=8$ h by an authorisation — fires only when the authorisation has expired. Each is exactly the kind of “property that contains time” a per-call classifier, and an unbounded past-time check, cannot express.

Multi-instance building actuation. To show the effect is not a single hand-picked pair, we author a 20-event smart-building trace (13 policy-relevant actions) exercising three past-time policies with multiple compliant and violating instances each (envelope-plus-maintenance-mode, a timed actuator dependency, and badge-authorised access; formulas in Appendix A). MonPoly flags exactly the 6 history-violating instances and clears all 7

Table 5: Benchmark readiness for history-dependent enforcement: fraction of runs/records exposing each field. Real timestamps, structured state transitions, and reversibility typing are absent from all three (call-order only).

Field	AgentDojo	STAC	R-Judge
explicit approval	0%	1.2%	0.2%
recipient/payee id	38.8%	58.4%	5.8%
object id	36.4%	34.2%	47.3%
user confirmation	4.6%	16.4%	0%
real timestamps	none	none	none
structured state	none	none	none
reversibility typing	none	none	none

compliant ones; one violation is a purely *timed* miss — an enabling damper_open that fell 4 h before a 3 h window — so the same compressor start is a violation only by timing.

R-Judge IoT (real data, small — cannot be enlarged). The R-Judge IoT subset (smart home, traffic dispatch, phone) is *all the labelled IoT agent records the benchmark ships*: 30 records (19 unsafe, 11 safe), so we report it as illustrative. The generic obligations yield 31.6% detection (6/19) and 27.3% FPR (3/11) with wide Wilson intervals. It does contain relational cases: in one unsafe record the agent grants a dog-walker *permanent* smart-lock access without consent. Encoding it under a parametric consent obligation $\square \forall g, p. (\text{grant}(g, p) \wedge p=1 \wedge \neg \diamond \text{consent}())$, MonPoly fires on the permanent grant and clears once a prior consent is present — a verdict flip on real IoT data the generic “grant” typing (which never fires here) cannot make.

10 THE BENCHMARK GAP: RECORDED TRACES LACK THE FIELDS HISTORY-DEPENDENT ENFORCEMENT NEEDS

The generic obligations reduce to typed-action detection not because of MFOTL but because of the data: the recorded traces omit the fields a history-dependent policy quantifies over. A readiness scorecard over the three benchmarks (Table 5) makes this concrete. Explicit *approvals* are almost absent — 0/6899 (0%) AgentDojo events, 6/483 (1.2%) STAC chains, 1/571 (0.2%) R-Judge records. Recipient/payee identity is recoverable more often (38.8%/58.4%/5.8%) and object identity sometimes (36.4%/34.2%/47.3%), but user-confirmation events are rare (4.6%/16.4%/0%). Above all, *none* of the three carries real timestamps (only call order), structured state transitions, or reversibility typing. The metric and stateful policies that motivate temporal RV (§9) thus cannot be evaluated on these corpora as shipped — we authored timed logs to exercise them.

A *minimal trace schema*. We therefore propose a minimal enforcement-ready schema — twelve fields per event: timestamp, actor, tool, action, object, recipient, source, approval_id, session_id, state_before, state_after, trust_domain. Current benchmarks supply the middle group (tool/action, partially object/recipient) but none of wall-clock timestamp, approval_id,

session_id, the state pair, or trust_domain — the fields the timed policies (§9) and the hardened provenance spec (§8) require. Cheap to emit but absent everywhere, they are the concrete obstacle to evaluating history-dependent agent enforcement.

11 DISCUSSION AND CONCLUSION

Positioning. Table 6 (appendix) contrasts the paradigms as demonstrated in the cited work and its evaluated configurations, not as a fundamental-impossibility claim. AgentDojo [6], AgentBench [12] and ToolEmu [13] evaluate agents; the guardrails they benchmark are, as reported, per-call or rule-based. AgentSpec [15] has a formal rule semantics and some history-aware rules (○), so a determined encoding could reach further than the cells show. Formal RV combines trace-level scope, formal temporal semantics, data-parametric provenance, and offline evaluability in one declarative substrate — not that the others provably cannot.

Limitations. We measure firing on recorded trajectories, not live-agent prevention; an in-loop shield addresses a different question (agent adaptation after enforcement) outside our scope. Generic specs have a high BFR (31.7%), and the parametric refinements that reduce it are evaluated for banking and workspace only. STAC lacks preconditions (6/483 chains), so its generic obligations reduce to typed-action detection; the metric power of MFOTL is exercised in the CPS/IoT study instead. Keyword typing is coarse, and a semantic residual ($\approx 29\%$ of STAC, more of R-Judge) needs a content checker structural RV does not provide. Finally, the timed policies run the real MonPoly engine but on authored logs that isolate the timing dependence — establishing expressiveness, not field prevalence.

Conclusion. Offline on real agent-attack corpora, MonPoly flags 71.8% of STAC chains and detects 61.6% of GPT-4o’s successful AgentDojo attacks at 31.7% BFR — numbers that mostly reflect risky-action typing, since these corpora rarely contain the preconditions temporal logic is for. Where data-parametric and timed reasoning applies, MFOTL states and MonPoly checks conditions a stateless per-call predicate cannot. Three completed results anchor the security case: a concrete earliest-warning margin (preventive on 51.3% of successful AgentDojo attacks, median lead two tool-calls); the provenance predicate behind the parametric precision is itself an attack surface, flipped 94–99% by one planted structured line, which we answer with a hardened trusted-provenance specification; and a readiness scorecard showing current benchmarks lack the timestamps, structured state, reversibility typing and approvals history-dependent enforcement needs. MFOTL/MonPoly is thus a reusable, auditable substrate for quantified, timed agent-trace policies, complementary to rule-based enforcement and measurable offline on the corpora the community already ships.

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A REPRODUCIBILITY

This appendix is excluded from the page limit. All scripts are pure-stlib Python in agent-rv-impl/; the datasets are the public releases below.

Datasets. STAC (arXiv:2509.25624, amazon-science/Multi-TurnAgentAttack, 483 chains, file STAC_benchmark_data.json); AgentDojo (ethz-spylab/agentdojo, recorded runs/<model>/<suite>/..., $\approx 36.7k$ runs; we use gpt-4o-2024-05-13 in detail and, for the cross-model analysis, the 22 base (no-defence) model directories the release ships, plus their defence variants where pooled — 28 directories with a banking suite and 29 with a workspace suite); R-Judge (Lordog/R-Judge, data/, 571 records; the IoT subset data/IoT/{household,trafficdispatch,phone_iot}.json is all 30 IoT records the benchmark ships and cannot be enlarged offline).

Keyword patterns (typing). A tool matches a class iff its lower-cased name contains any listed substring. *DESTRUCTIVE*: delete, remove, cancel, clear, wipe, drop, purge, erase, revoke, uninstall. *TEARDOWN*: autodelete, disable, stop, unlock, deactivate, turn_off, bypass, whitelist, allowlist, add_user_autodelete. *FINANCIAL*: send_money, transaction, transfer, payment, pay_, book_, buy, purchase, zelle, venmo, wire, invoice, checkout. *EXPORT*: send_email, share, export, upload, post_, publish, forward, sendmessage, send_message, send_sms, tweet. *ACTUATION*: set_, adjust, trafficlight, unlock_door, execute, terminal, medication, open_, close_, activate, reboot, restart, deploy, write_, update_, configure, install. *APPROVAL (precondition)*: confirm, approve, consent, authorize, authorise, permission, verify, grant, two_factor, 2fa, otp, request_approval, ack, sign_off, double_check.

Argument normalisation. Tool arguments are parsed from JSON to a dict (malformed strings kept under a raw key). The known_payee and known_contact predicates are *not* hand-curated: they are emitted by the same scripted canonicaliser via a fixed regular expression over structured recipient/sender fields, and the regexes were frozen *before* we inspected any attacker IBAN/address or any security/utility outcome. For the payee obligation a recipient is a *known payee* only if it matched $(?:recipient|sender):s*([A-Z]{2}[0-9A-Z]{6,})$ in prior message text (a structured IBAN field, not free text). For the external-recipient obligation an address is a *known contact* only if it appeared under a directory key $(?:from|sender|recipients?|participants?|email|shared_with|owner):$ in a prior *tool* result. So a payee/contact becomes “known” purely by the scripted rule, independent of whether it is later an attack address.

Approval/consent recognition. An approval precedes step i iff some call at $j < i$ (same trace, same STAC case) matches the APPROVAL patterns. An obligation fires on step i iff the tool matches its class patterns *and* no approval precedes i .

Timestamps and multiple firings. Each canonical event becomes one MonPoly timepoint @ t ; for STAC the class and approval predicates are parameterised by the case id so ONCE only matches within a case. Only timepoints that carry at least one typed predicate are emitted: a canonicalised step whose tool matches *no* obligation class (a read-only or out-of-template action) produces no event, so the log holds 357 predicate-bearing timepoints across the 483 cases, and cases whose final step matches no class emit nothing and cannot fire (counted as a miss). Because the STAC obligations use an unbounded ONCE[0, 10⁶], dropping predicate-free timepoints is sound — an unbounded past-time operator is insensitive to the number of intervening empty timepoints, only to whether a matching predicate occurred earlier in the case. For the *timed* policies of §9 this is different: there the integer timestamp is the event’s clock time (in hours), timepoints are spaced by real elapsed time, and the metric windows (ONCE[0, W]) depend on those gaps, so we keep the actual times rather than a dense index. Per-obligation counts are non-exclusive; a case/run counts as detected if ≥ 1 obligation fires.

Full MFOTL obligations (monitored violation form). With $\blacklozenge$ once-in-the-past:

O1 : destructive(a) $\wedge$ $\neg \blacklozenge$ approved(a)
O2 : teardown(a) $\wedge$ $\neg \blacklozenge$ approved(a)
O3 : financial(a) $\wedge$ $\neg \blacklozenge$ approved(a)
O4 : export(a) $\wedge$ $\neg \blacklozenge$ approved(a)
O5 : actuation(a) $\wedge$ $\neg \blacklozenge$ approved(a)

The STAC export uses the disjunction handed verbatim to MonPoly: (destructive(c) OR teardown(c) OR financial(c) OR export(c) OR actuation(c)) AND NOT (ONCE[0,1000000] approval(c)). The two parametric obligations are $\Box \forall p.(\text{transfer}(p) \wedge \neg \blacklozenge \text{known_payee}(p))$ and $\Box \forall r.(\text{send}(r) \wedge \neg \blacklozenge \text{known_contact}(r))$. The multi-instance building policies, run verbatim by MonPoly, are:

P1: setpoint(z,v) AND $v > 28$ AND NOT ((NOT maint_off()) SINCE maint_on())
P2: compressor_start(z) AND NOT (ONCE[0,3] damper_open(z))
P3: door_unlock(d) AND NOT (ONCE[0,3] badge_auth(d))
lock: grant_access(g,p) AND $p = 1$ AND NOT (ONCE consent())

The 20-event building log yields firings at timepoints {4,5,9,14,19,22} (6 violations; the P2 firing at $t=14$ is the timed miss, a damper_open at $t=10$ outside the 3-unit window) and no firing on the 7 compliant instances.

Timed (metric) policies of §9. These use hour-valued timestamps; formula, signature and log are run verbatim by MonPoly:

T1: export(o) AND NOT (ONCE[0,24] consent(o))
log: @0 consent(1) @12 export(1) @48 export(1) @50 consent(1) @60 export(1)
 $\Rightarrow$ fires only @48 (consent 48h stale)
T2: ($c \leftarrow$ CNT s ; d (ONCE[0,6] actuate(d,s))) AND $c >= 3$
log: @0 actuate(7,1) @2 actuate(7,2) @5 actuate(7,3) @30.. @33.. @40..
 $\Rightarrow$ fires only @5 (3rd actuation within 6h; spaced calls never fire)
T3: execute(x) AND NOT (ONCE[0,8] request(x))
log: @0 request(5) @6 execute(5) @40 execute(5) @44 request(5) @48 execute(5)
 $\Rightarrow$ fires only @40 (authorisation 40h old)

Each timed verdict flips on the *same* action purely by elapsed time; T2 uses MonPoly’s metric CNT aggregation over a sliding [0, 6] window.

MonPoly version and invocation. The engine is MonPoly, version string MonPoly (development build) (from monpoly -version), at /usr/local/bin/monpoly inside Docker image rv-fabric-impl-backend (image id sha256:6e22367466a8). It is invoked as monpoly -sig < s >.sig -formula < f >.mfotl -log < l >.log; all formulas above pass monpoly -check as monitorable. On the STAC export it fires on exactly 347 timepoints, matching the reference monitor.

Artifact availability. All evaluation scripts are pure-stdlib Python under agent-rv-impl/; together with the MonPoly signature/log/formula files for the STAC, building, IoT and timed studies they will be released as supplementary material (and, on acceptance, an anonymous public repository), so the full pipeline is reproducible against the public dataset releases named above.

Table 8: Per-suite replay (GPT-4o, no defence): AgentDojo attack-success, RV detection on successful attacks, and RV BFR on benign runs.

Suite	ASR	RV detect	BFR
banking	40.1%	97.7%	95.8%
slack	60.9%	40.3%	11.5%
travel	9.3%	16.5%	11.1%
workspace	19.9%	62.9%	21.7%

AgentDojo selection and unflagged successful-attack fraction. Runs are read by recursive glob of runs/<model>/**/*.*.json. A run is *attacked* iff `attack_type` is set and `≠none` and `injection_task_id` is present; *benign* iff neither is present; runs with an empty/unparseable tool-call chain are skipped (96 for GPT-4o $\Rightarrow$ 6680 attacked, 123 benign). `security==True` marks a successful attack, `utility==True` a completed task. For reference we report the *unflagged successful-attack fraction*, ($\text{successful} \wedge \neg \text{RV-fired}$)/attacked = $(1931 - 1190)/6680 = 11.1\%$; this is a descriptive count of successful attacks the monitor did not flag, *not* a measured attack-success rate under an in-loop shield (which would require re-running the agent).

B EXTENDED RESULTS AND ROBUSTNESS

Table 6: Paradigm comparison, per each system’s evaluated configuration (not a fundamental-capability claim). TL = trace-level; FS = formal temporal semantics; PAR = parametric (quantifies over trace data); OFF = evaluable offline on recorded runs. $\circ$ = partial / achievable with bespoke stateful code.

	TL	FS	PAR	OFF
AgentDojo defences [6]	–	–	–	✓
AgentSpec [15]	$\circ$	✓	$\circ$	–
Progent [14]	$\circ$	–	–	–
GuardAgent [16]	–	–	–	–
STAC [11] (benchmark)	✓	–	–	✓
Formal RV / MonPoly (ours)	✓	✓	✓	✓

Per-obligation ablation (AgentDojo, GPT-4o). Table 7 breaks the union result down by obligation, with each obligation’s positive likelihood ratio LR^+ (detection / BFR). Export is the single most discriminating obligation ($\text{LR}^+ = 2.58$) and teardown never fires for GPT-4o (these suites have no protection-teardown tools).

Table 7: Per-obligation detection on successful attacks and BFR (AgentDojo, GPT-4o), with unique flagged successful attacks and $\text{LR}^+ = \text{detection/BFR}$. Union is the headline any-obligation result.

Obligation	detect %	BFR %	unique	LR^+
O1 destructive	7.5	3.3	105	2.29
O2 teardown	0	0	0	–
O3 financial	28.0	17.1	245	1.64
O4 export	27.3	10.6	488	2.58
O5 actuation	16.2	6.5	18	2.49
union (any)	61.6	31.7	–	1.94

Cross-model sign test (parametric payee, banking). Beyond the pooled McNemar test (§6), we check the direction of the parametric effect per model. Across the 22 base (no-defence) model directories with a banking suite, the parametric payee obligation *decreases* benign firing in 20, leaves it unchanged in 2, and increases it in 0; the median per-directory reduction is 8.0 percentage points (IQR [4, 17], min 0, max 24). An exact two-sided sign test on the 20-vs-0 split gives $p = 1.9 \times 10^{-6}$: the reduction direction is consistent across models, not an artefact of pooling. Over the same 22 directories, generic detection ranges 40.9–83.3% and BFR 17.6–37.4%.

Table 9: Canonicalisation typing vs. 54 hand-labelled tools. Per-class multi-label precision/recall (does the typing include the gold class?), micro-average, and single-primary exact-match.

Class	P	R	(tp/fp/fn)
O1 destructive	0.78	1.00	7/2/0
O2 teardown	1.00	1.00	3/0/0
O3 financial	0.78	0.78	7/2/2
O4 export	1.00	0.67	4/0/2
O5 actuation	0.86	0.86	6/1/1
micro-average	0.84	0.84	F1 0.84
single-primary exact-match:	45/54 = 83.3%		

Canonicalisation sensitivity. Table 10 re-runs the AgentDojo and STAC replays under four tool $\rightarrow$ class mappings. Correcting the known typing errors raises detection and BFR together (the frozen keyword typing is conservative); the conservative mapping leaves AgentDojo unchanged but lowers STAC.

Table 10: Sensitivity of the headline rates to the tool $\rightarrow$ class mapping. AD = AgentDojo (GPT-4o); STAC detection is final-step.

Mapping	AD detect %	AD BFR %	STAC detect %
original	61.6	31.7	71.8
corrected	71.1	42.3	73.1
conservative	61.6	31.7	61.7
expanded	71.2	45.5	73.1

MonPoly runtime and scale-up soundness. Table 11 reports throughput and peak resident memory for the STAC log at 1 $\times$, 10 $\times$ and 50 $\times$ replication and for an AgentDojo-derived log. Throughput stays near 0.4–0.8M events/s with flat 8–14 MB RSS, and parametric/timed formulas cost about the same as generic ones. Replicating the STAC log 10 $\times$ yields exactly $3470 = 10 \times 347$ firings, a soundness check on scale-up.

Table 11: MonPoly throughput and peak RSS. STAC $\times k$ is the STAC log replicated k times; firings scale exactly ($10 \times 347 = 3470$).

Log	throughput (events/s)	peak RSS
STAC $\times 1$	$\approx 0.4\text{M}$	8–14 MB
STAC $\times 10$	452k	8–14 MB
STAC $\times 50$	429k	8–14 MB
AgentDojo-derived	$\approx 0.8\text{M}$	8–14 MB

Synthetic corpus (AgentTrace-T) and metamorphic tests. To exercise the specifications where the public corpora lack the needed fields, we generated *AgentTrace-T*: 1400 traces, 200 per policy across P1–P3, T1–T3 and the IoT-lock policy, in four categories (compliant, violating, boundary, and adversarially-ambiguous), roughly 100 compliant and 100 violating each. MonPoly achieves 100% verdict accuracy, 100% parameter-binding accuracy, and 100% on the boundary cases. A metamorphic test suite then applies six semantics-preserving or semantics-changing transformations — moving an enabler across the window boundary, changing an id, reordering, inserting an irrelevant event (verdict must not change), duplicating an event (must not change), and boundary shifts — and MonPoly gives the expected verdict on 780/780 (100%).

Combined parametric ablation. Applying *both* parametric obligations together, the banking+workspace pooled rates move from generic 80.8% detection / 46.5% BFR to 79.8% / 43.7%, and the four-suite overall rates from generic 61.5% / 31.5% to both-parametric 60.9% / 29.8%. Run-level BFR barely moves because benign runs

also trip *other* generic obligations; the parametric win is at the *action slot* — the export slot BFR falls 9.9% $\rightarrow$ 7.0% and the *send_money* slot 44% $\rightarrow$ 24% — which is where the provenance conjunct applies.

Provenance poisoning (detail). The poisoning experiment (§8) injects, per flagged run, one line naming the attacker recipient in four framings and re-evaluates the parametric obligation. Banking payee (403 flagged successful-attack runs): structured 379/403 = 94.0% flip, quoted/forwarded 94.0%, free-text 0%, reformatted-IBAN 0%. Workspace recipient (203 flagged): structured 201/203 = 99.0%, quoted 99.0%, free-text 0%, reformatted 0%.

Scripts. `stac_rv.py`, `stac_to_monpoly.py`, `agentdojo_rv.py`, `agentdojo_scenarios.py`, `agentdojo_param.py` (payee), `agentdojo_param2.py` (external recipient), `agentdojo_param_pooled.py` (pooled cross-model McNemar — produces the headline p -values), `cross_model.py` (per-directory cross-model sign test), `first_firing.py` (earliest-warning), `combined_ablation.py` and `per_oblig_ablation.py` (ablations), `risk_enrichment.py` (LR⁺ and balanced accuracy), `canon_sensitivity.py` (four-mapping sensitivity), `perturb_provenance.py` (poisoning), `readiness_scorecard.py` (benchmark scorecard), `corpus_gen.py` (AgentTrace-T), `metamorphic.py` (metamorphic tests), `mp_time.py` (MonPoly runtime), `canon_eval.py`, `wilson_ci.py`, `mcnemar.py` (single-model paired-benign, the inconclusive per-model result), `rjudge_rv.py`, and the MonPoly `.sig/.mfotl/.log` files for the STAC, building, IoT and timed studies.